



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-082
HEARTSEASE-MAYO DUNBAR DP
18 MW Capacity / 18 MW Energy**

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Dominion.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

3 General

The Interconnection Customer (IC) has proposed an uprate to a planned solar generating facility located in Edgecombe, North Carolina. This project is an increase to the Interconnection Customer's AC1-034 project, which will share the same point of interconnection. The AF1-082 queue position is an 18 MW storage uprate (18 MW Capacity uprate) to the previous project. The total installed facilities will have a capability of 93 MW with 60.75 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this uprate project is December 31, 2020. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-082
Project Name	HEARTSEASE-MAYO DUNBAR DP
State	North Carolina
County	Edgecombe
Transmission Owner	Dominion
MFO	93
MWE	18
MWC	18
Fuel	Storage
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-082 will interconnect with the Dominion transmission system at the new three breaker ring bus AC1-034 substation tapping the Heartsease to Mayo Dunbar DP 115 kV line.

5 Cost Summary

The AF1-082 project will utilize the Interconnection Facilities being built for the AC1-034 project.

The AF1-082 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 0
Allocation towards System Network Upgrade Costs*	\$ 0
Total Costs	\$ 0

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start

with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AF1-082. The project was evaluated as an 18 MW Capacity (18.0 MW energy) injection at the planned AC1-034 substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2023 AF1 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically, in Planning Studies, NERC Planning Event 3 and 6 Contingency Conditions (Loss of generator, transmission circuit, transformer, shunt device, or Single Pole of a DC line followed by the loss of a generator, transmission circuit, transformer, shunt device or single pole of a DC line) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

Note that the ITO findings were made from a conceptual review of this project and cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein

reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

7 Schedule

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

8 Transmission Owner Analysis

8.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2023 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no further deficiencies were identified.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion’s “Dominion Energy Electric Transmission Generator Interconnection Requirements” documented in Dominion’s Facility Interconnection Requirements “Exhibit C” located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC’s GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at:
<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Interconnected Transmission Owner Requirements

See Section 3.4.6 “Metering and telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

11 Summer Peak Analysis

The Queue Project AF1-082 was evaluated as a 18.0 MW (Capacity 18.0 MW) uprate to AC1-034 which is an injection tapping the Mayo Dunbar to Heartsease DP 115 kV line in the Dominion area. Project AF1-082 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-082 was studied with a commercial probability of 100%. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
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ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
89913747	313845	6HATHAWAY	230.0	DVP	314591	6NASH	230.0	DVP	1	DVP_P1-2: LN 2058	single	449.32	99.29	100.02	AC	3.39

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42863883	314554	3BTLEBR O	115.0	DVP	304223	3ROCKYMT11 5T	115.0	CPL	1	DVP_P4-2: 2058T2167	breaker	93.0	218.81	225.27	AC	6.31
42864014	314554	3BTLEBR O	115.0	DVP	304223	3ROCKYMT11 5T	115.0	CPL	1	DVP_P1-2: LN 2058	single	93.0	154.56	160.83	AC	6.19
42864015	314554	3BTLEBR O	115.0	DVP	304223	3ROCKYMT11 5T	115.0	CPL	1	DVP_P1-3: 6HATHAWAY-TX#1	single	93.0	141.81	147.9	AC	6.0
42864018	314554	3BTLEBR O	115.0	DVP	304223	3ROCKYMT11 5T	115.0	CPL	1	Base Case	single	93.0	112.72	117.03	AC	4.31
42864276	314554	3BTLEBR O	115.0	DVP	304223	3ROCKYMT11 5T	115.0	CPL	1	DVP_P7-1: LN 2058-2181	tower	93.0	445.63	453.18	AC	7.38

11.4 Steady-State Voltage Requirements

To be determined

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
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ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
43051377	313719	3CHESTNUT	115.0	DVP	314623	3WITAKRS	115.0	DVP	1	DVP_P1-2: LN 2056-A	operation	165.44	128.1	129.81	AC	2.84
42864091	313845	6HATHAWAY	230.0	DVP	304222	6ROCKYMT30T	230.0	CPL	1	DVP_P1-2: LN 2181	operation	478.0	117.75	118.78	AC	4.87
46053081	313854	3CONSLDL	115.0	DVP	314554	3BTLEBRO	115.0	DVP	1	DVP_P1-2: LN 126	operation	165.44	132.0	133.12	AC	1.87
46053087	313854	3CONSLDL	115.0	DVP	314554	3BTLEBRO	115.0	DVP	1	Base Case	operation	165.44	106.05	107.42	AC	2.3
43051511	313858	3SO JUSTICE	115.0	DVP	314577	3COX DP	115.0	DVP	1	DVP_P1-2: LN 2056-A	operation	165.44	101.29	102.98	AC	2.84
42864009	314554	3BTLEBRO	115.0	DVP	304223	3ROCKYMT115T	115.0	CPL	1	DVP_P1-2: LN 2058	operation	93.0	213.92	220.27	AC	6.19
42864013	314554	3BTLEBRO	115.0	DVP	304223	3ROCKYMT115T	115.0	CPL	1	Base Case	operation	93.0	155.35	159.73	AC	4.31
43051450	314556	3BENSON	115.0	DVP	313844	3HATHAWAY	115.0	DVP	1	Base Case	operation	246.28	110.29	117.36	AC	17.99
46053073	314623	3WITAKRS	115.0	DVP	313854	3CONSLDL	115.0	DVP	1	DVP_P1-2: LN 126	operation	165.44	132.0	133.13	AC	1.87
46053079	314623	3WITAKRS	115.0	DVP	313854	3CONSLDL	115.0	DVP	1	Base Case	operation	165.44	106.05	107.42	AC	2.3
43051427	924150	AB2-059 TAP	115.0	DVP	314556	3BENSON	115.0	DVP	1	Base Case	operation	246.28	112.99	120.06	AC	17.99

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-082	Upgrade Number
42864276,42863883,42864014,42864015,42864018	2	3BTLEBRO 115.0 kV - 3ROCKYMT115T 115.0 kV Ckt 1	<u>DVP</u> n6118: Upgrading the breaker leads at DVP's terminal will bring the rating to 239/239/239 MVA (Limited by terminal equipment at Rocky Mount) Dominion End Ratings: 438/478/581 MVA Project Type : FAC Cost : \$100,000 Time Estimate : 30-36 Months New Ratings: Rate A: 239 MVA Rate B: 239 MVA Rate C: 239 MVA Queue Project AF1-082 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-082 could receive cost allocation. Note 2: Although Queue Project AF1-082 may not have cost responsibility for this upgrade, Queue Project AF1-082 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-082 comes into service prior to completion of the upgrade, Queue Project AF1-082 will need an interim study. n6618: 1. Split the 155 kV Bus at Hathaway into two separate buses with a 115 kV Line on each bus. 2. Rebuild Line #55 (Tarboro – Anaconda) and close the tie switch between Line 55 & 80. 3. Line #1001 is opened at Battleboro thus making Line #1001 radial from Chestnut Substation Project Type : CON Cost : \$6,900,000 Time Estimate : 20 Months The upgrade will need to be tested during the Facilities Study phase to ensure it does not cause any additional violations. Queue Project AF1-082 presently does not receive cost allocation for this upgrade. Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-082 could receive cost allocation. Note 2: Although Queue Project AF1-082 may not have cost responsibility for this upgrade, Queue Project AF1-082 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-082 comes into service prior to completion of the upgrade, Queue Project AF1-082 will need an interim study. <u>DEP</u> See Next Page	See Next Page	See Next Page	n6118, n6618

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-082	Upgrade Number
42864276,42863883,42864014,42864015,42864018	2	3BTLEBRO 115.0 kV - 3ROCKYMT115T 115.0 kV Ckt 1	DEP (cont'd) Note 1: Duke Energy/Progress Portion: A potential constraint was identified by PJM on the Duke Energy/Progress (DEP) portion of the Battleboro- Rocky Mountain 115 kV line. The Queue Project AF1-082 may be subject to operational restriction if real-time system reliability issues occur. Additionally, if a baseline Network Upgrade project is identified on the Battleboro- Rocky Mountain 115 kV line prior to the execution of the Queue Project's final agreements, the Queue Project may require this upgrade to be in-service to be deliverable to the PJM system. If Queue Project AF1-082 comes into service prior to completion of the baseline Network Upgrade, Queue Project AF1-082 will need an interim deliverability study Note 2: There are no mitigations currently planned for the DEP portion of this overload. Though, the following upgrade has been scoped by DEP to increase the DEP End Rating to 301/301/301. Note 3: The existing DEP rating on Battleboro – Rocky Mountain 115 kV line is 164/164/164 MVA. Scope: Reconductor the circuit to 1590 ACSR will bring the rating to 185/205/214 MVA. Upgrading the terminal equipment at Rocky Mount will bring the rating to 301/301/301 MVA (Limited by 1590 ACSR conductor). Project Type: FAC Cost: TBD Time Estimate: 36 Months	\$7,000,000 Excludes DEP Upgrade Costs	\$0	n6118, n6618

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-082	Upgrade Number
89913747	1	6HATHAWAY 230.0 kV - 6NASH 230.0 kV Ckt 1	DVP n6442: Rebuild 0.76 miles of 230 kV Line 2181 from Hathaway to Nash with 2-636 ACSR. Project Type : FAC Cost : \$1,900,000 Time Estimate : 30-36 Months New Ratings: Rate A: 1047 MVA Rate B: 1047 MVA Rate C: 1204 MVA Queue Project AF1-082 presently does not receive cost allocation for this upgrade. Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-082 could receive cost allocation. Note 2: Although Queue Project AF1-082 may not have cost responsibility for this upgrade, Queue Project AF1-082 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-082 comes into service prior to completion of the upgrade, Queue Project AF1-082 will need an interim study.	\$1,900,000	\$0	n6442
			TOTAL COST	\$8,900,000	\$0	

Note: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test

rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
89913747	313845	6HATHAWAY	DVP	314591	6NASH	DVP	1	DVP_P1-2: LN 2058	single	449.32	99.29	100.02	AC	3.39

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
314435	6SAPONY	0.1839	80/20	0.1839
314572	3EMPORIA	0.0508	80/20	0.0508
314574	6EVERETS	0.2392	80/20	0.2392
314582	3KELFORD	0.3543	80/20	0.3543
314589	3MURPHYS	0.0609	80/20	0.0609
314623	3WITAKRS	0.2272	80/20	0.2272
314704	3LAWRENC	0.1730	80/20	0.1730
315115	1S HAMPT1	0.7880	80/20	0.7880
315126	1ROARAP2	1.1760	80/20	1.1760
315128	1ROARAP4	1.1223	80/20	1.1223
315131	1EDGECEMA (Deactivation : 22/04/2019)	26.1916	80/20	26.1916
315132	1EDGECEMB (Deactivation : 22/04/2019)	26.1916	80/20	26.1916
315136	1ROSEMG1	3.0631	80/20	3.0631
315137	1ROSEMS1	1.8995	80/20	1.8995
315138	1ROSEMG2	1.4356	80/20	1.4356
315139	1GASTONA	3.7732	80/20	3.7732
315141	1GASTONB	3.7732	80/20	3.7732
315292	1DOMTR78	0.9001	80/20	0.9001
315294	1DOMTR10	6.5863	80/20	6.5863
315601	1CONETOE2SOL	2.8887	80/20	2.8887
315602	1HOLLOMANSOL	1.0866	80/20	1.0866
315605	6W1-029WIND	0.2526	80/20	0.2526
315606	3AA2-053SOLA	1.2006	80/20	1.2006
315607	3AA1-063SOLA	1.0815	80/20	1.0815
315608	3AA2-088SOLA	0.6658	80/20	0.6658
315611	6Z1-036WIND	2.4699	80/20	2.4699
315612	3AA2-057SOLA	1.1723	80/20	1.1723
315614	AA2-178 C	0.8178	80/20	0.8178
920591	AA2-165 C	0.1390	80/20	0.1390
920671	AA2-174 C OP	0.0550	80/20	0.0550
922922	AB1-081 C OP	10.5437	80/20	10.5437
923262	AB1-132 C OP (Suspended)	16.2683	80/20	16.2683
923572	AB1-173 C OP	1.6317	80/20	1.6317
923582	AB1-173AC OP	1.6317	80/20	1.6317
923801	AB2-015 C OP	3.9755	80/20	3.9755
923911	AB2-031 C O1	1.6197	80/20	1.6197
923991	AB2-040 C O1 (Suspended)	5.3183	80/20	5.3183
924151	AB2-059 C OP	2.2009	80/20	2.2009

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
924501	AB2-099 C (Suspended)	0.4009	80/20	0.4009
924511	AB2-100 C	1.5313	80/20	1.5313
925121	AB2-169 C	0.7615	80/20	0.7615
925171	AB2-174 C O1	0.8813	80/20	0.8813
925591	AC1-034 C	8.0490	80/20	8.0490
925781	AC1-054 C O1	2.8124	80/20	2.8124
926071	AC1-086 C	23.9570	80/20	23.9570
926201	AC1-098 C	5.1354	80/20	5.1354
926211	AC1-099 C	1.7209	80/20	1.7209
927021	AC1-189 C	12.7226	80/20	12.7226
927141	AC1-208 C	7.7881	80/20	7.7881
932631	AC2-084 C	7.3207	80/20	7.3207
933991	AD1-023 C	7.7071	80/20	7.7071
934331	AD1-057 C O1	20.9131	80/20	20.9131
934521	AD1-076 C	30.0752	80/20	30.0752
936401	AD2-051 C O1	6.1077	80/20	6.1077
938171	AE1-026 C O1	19.4268	80/20	19.4268
938221	AE1-035 C	1.5338	80/20	1.5338
938771	AE1-103 C O1	1.5767	80/20	1.5767
940481	AE2-033 C	7.2980	80/20	7.2980
940491	AE2-034 C	3.4629	80/20	3.4629
940541	AE2-040 O1	1.1816	80/20	1.1816
940571	AE2-044 C	5.2718	80/20	5.2718
941501	AE2-147 C	4.7547	80/20	4.7547
941541	AE2-151 C	0.7619	80/20	0.7619
942471	AE2-260 C O1	10.0034	80/20	10.0034
942851	AE2-304 C	0.2091	80/20	0.2091
943171	AE2-346 C	0.9621	80/20	0.9621
943911	AF1-059	6.8240	80/20	6.8240
944141	AF1-082	3.3890	80/20	3.3890
944871	AF1-152 C	1.5849	80/20	1.5849
945711	AF1-236 C O1	37.9105	80/20	37.9105
946011	AF1-266	6.1090	80/20	6.1090
946281	AF1-292 C	0.9980	80/20	0.9980
NEWTON	NEWTON	1.3689	Confirmed LTF	1.3689
FARMERCITY	FARMERCITY	0.0785	Confirmed LTF	0.0785
G-007A	G-007A	0.8535	Confirmed LTF	0.8535
VFT	VFT	2.2639	Confirmed LTF	2.2639
PRAIRIE	PRAIRIE	3.6937	Confirmed LTF	3.6937
COFFEEN	COFFEEN	0.2558	Confirmed LTF	0.2558
CHEOAH	CHEOAH	1.1141	Confirmed LTF	1.1141
EDWARDS	EDWARDS	0.4134	Confirmed LTF	0.4134
TILTON	TILTON	0.7314	Confirmed LTF	0.7314
GIBSON	GIBSON	0.6618	Confirmed LTF	0.6618
CALDERWOOD	CALDERWOOD	1.0864	Confirmed LTF	1.0864
BLUEG	BLUEG	2.0433	Confirmed LTF	2.0433
TRIMBLE	TRIMBLE	0.6489	Confirmed LTF	0.6489
CATAWBA	CATAWBA	1.1725	Confirmed LTF	1.1725

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42864276	314554	3BTLEBRO	DVP	304223	3ROCKYMT115T	CPL	1	DVP_P7-1: LN 2058-2181	tower	93.0	445.63	453.18	AC	7.38

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
314541	3WATKINS	0.2104	Adder	0.25
314582	3KELFORD	0.3985	50/50	0.3985
314623	3WITAKRS	0.5684	50/50	0.5684
315131	1EDGECEMA (Deactivation : 22/04/2019)	12.1440	50/50	12.1440
315132	1EDGECEMB (Deactivation : 22/04/2019)	12.1440	50/50	12.1440
315136	1ROSEMG1	1.6358	50/50	1.6358
315137	1ROSEMS1	1.0143	50/50	1.0143
315138	1ROSEMG2	0.7666	50/50	0.7666
315139	1GASTONA	2.1022	50/50	2.1022
315141	1GASTONB	2.1022	50/50	2.1022
315601	1CONETOE2SOL	1.3631	50/50	1.3631
315612	3AA2-057SOLA	2.6249	50/50	2.6249
900672	V4-068 E	0.1238	Adder	0.15
917332	Z2-043 E	0.9783	50/50	0.9783
917342	Z2-044 E	1.3755	50/50	1.3755
917512	Z2-088 E OP1	3.2983	50/50	3.2983
918492	AA1-063AE OP	1.9435	Adder	2.29
918512	AA1-065 E OP	1.4302	Adder	1.68
918532	AA1-067 E	0.2465	Adder	0.29
919692	AA2-053 E OP	1.9785	Adder	2.33
919702	AA2-057 E OP	6.9959	50/50	6.9959
920042	AA2-088 E OP	4.0157	Adder	4.72
920591	AA2-165 C	0.3112	50/50	0.3112
920592	AA2-165 E	0.9615	50/50	0.9615
920672	AA2-174 E OP	0.2286	Adder	0.27
922922	AB1-081 C OP	22.9678	50/50	22.9678
922923	AB1-081 E OP	9.8434	50/50	9.8434
923262	AB1-132 C OP (Suspended)	9.0636	50/50	9.0636
923263	AB1-132 E OP (Suspended)	3.8844	50/50	3.8844
923572	AB1-173 C OP	1.0076	Adder	1.19
923573	AB1-173 E OP	0.4702	Adder	0.55
923582	AB1-173AC OP	1.0076	Adder	1.19
923583	AB1-173AE OP	0.4702	Adder	0.55
923801	AB2-015 C OP	2.5105	Adder	2.95
923802	AB2-015 E OP	2.0586	Adder	2.42
923911	AB2-031 C O1	1.0002	Adder	1.18
923912	AB2-031 E O1	0.4926	Adder	0.58

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
923991	AB2-040 C O1 (Suspended)	3.2841	Adder	3.86
923992	AB2-040 E O1 (Suspended)	2.6870	Adder	3.16
924151	AB2-059 C OP	4.7943	50/50	4.7943
924152	AB2-059 E OP	13.9448	50/50	13.9448
924501	AB2-099 C (Suspended)	0.2636	Adder	0.31
924502	AB2-099 E (Suspended)	0.1130	Adder	0.13
924512	AB2-100 E	2.0925	Adder	2.46
925122	AB2-169 E	1.7713	Adder	2.08
925172	AB2-174 E O1	2.7080	Adder	3.19
925591	AC1-034 C	17.5335	50/50	17.5335
925592	AC1-034 E	13.2270	50/50	13.2270
926071	AC1-086 C	13.3472	50/50	13.3472
926072	AC1-086 E	6.0748	50/50	6.0748
926201	AC1-098 C	8.4829	50/50	8.4829
926202	AC1-098 E	5.0537	50/50	5.0537
926211	AC1-099 C	2.8427	50/50	2.8427
926212	AC1-099 E	1.6695	50/50	1.6695
927021	AC1-189 C	6.0689	50/50	6.0689
927022	AC1-189 E	3.0231	50/50	3.0231
927141	AC1-208 C	13.5209	50/50	13.5209
927142	AC1-208 E	6.0039	50/50	6.0039
932631	AC2-084 C	12.0927	50/50	12.0927
932632	AC2-084 E	5.9561	50/50	5.9561
933991	AD1-023 C	3.9006	Adder	4.59
933992	AD1-023 E	2.1235	Adder	2.5
934331	AD1-057 C O1	10.1813	50/50	10.1813
934332	AD1-057 E O1	5.4311	50/50	5.4311
936401	AD2-051 C O1	3.1690	Adder	3.73
936402	AD2-051 E O1	1.3607	Adder	1.6
938171	AE1-026 C O1	9.8964	Adder	11.64
938172	AE1-026 E O1	2.6109	Adder	3.07
938221	AE1-035 C	0.7986	Adder	0.94
938222	AE1-035 E	0.3933	Adder	0.46
940571	AE2-044 C	11.4839	50/50	11.4839
940572	AE2-044 E	4.9217	50/50	4.9217
941541	AE2-151 C	0.5010	Adder	0.59
941542	AE2-151 E	0.2698	Adder	0.32
942471	AE2-260 C O1	4.9509	Adder	5.82
942472	AE2-260 E O1	7.0222	Adder	8.26
943171	AE2-346 C	0.6326	Adder	0.74
943172	AE2-346 E	0.2711	Adder	0.32
944141	AF1-082	7.3825	50/50	7.3825
946281	AF1-292 C	0.5550	Adder	0.65
946282	AF1-292 E	0.3742	Adder	0.44
NEWTON	NEWTON	0.8671	Confirmed LTF	0.8671
FARMERCITY	FARMERCITY	0.0497	Confirmed LTF	0.0497
G-007A	G-007A	0.4795	Confirmed LTF	0.4795
VFT	VFT	1.2707	Confirmed LTF	1.2707
PRAIRIE	PRAIRIE	2.3402	Confirmed LTF	2.3402
COFFEEN	COFFEEN	0.1620	Confirmed LTF	0.1620
CHEOAH	CHEOAH	0.7042	Confirmed LTF	0.7042

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
EDWARDS	EDWARDS	0.2618	Confirmed LTF	0.2618
TILTON	TILTON	0.4631	Confirmed LTF	0.4631
GIBSON	GIBSON	0.4193	Confirmed LTF	0.4193
CALDERWOOD	CALDERWOOD	0.6869	Confirmed LTF	0.6869
BLUEG	BLUEG	1.2933	Confirmed LTF	1.2933
TRIMBLE	TRIMBLE	0.4107	Confirmed LTF	0.4107
CATAWBA	CATAWBA	0.7406	Confirmed LTF	0.7406

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-063	Huntsville (Cabin Creek) 69kV	Withdrawn
AA1-063A	Carolina-Seaboard 115kV	In Service
AA1-065	Earleys 230kV	In Service
AA1-067	Everetts 34.5kV	In Service
AA2-053	Carolina-Jackson 115kV	In Service
AA2-057	Hornertown-Whitakers 115kV	In Service
AA2-088	Boykins-Handsome 115kV	In Service
AA2-165	Hornertown-Whitakers 115kV	In Service
AA2-174	Carolina-Jackson 115kV	In Service
AA2-178	Mackeys 230kV	Under Construction
AB1-081	Anaconda-Mayo Dunbar 115kV	Under Construction
AB1-132	Thelma 230kV	Suspended
AB1-173	Brink-Trego 115kV	Engineering and Procurement
AB1-173A	Brink-Trego 115kV	Engineering and Procurement
AB2-015	Franklin 115kV	Engineering and Procurement
AB2-031	Brink-Trego 115kV	Engineering and Procurement
AB2-040	Brink 115kV	Suspended
AB2-059	Benson-Dunbar 115kV	Under Construction
AB2-099	Ahoskie 34.5kV	Suspended
AB2-100	Clubhouse-Lakeview 230kV	Under Construction
AB2-169	Pantago-Five Points 115kV	Partially in Service - Under Construction
AB2-174	Emporia-Trego 115kV	Under Construction
AC1-034	Heartsease DP - Mayo Dunbar 115kV	Active
AC1-054	Kerr Dam-Eatons Ferry 115 kV	Engineering and Procurement
AC1-086	Thelma 230kV	Active
AC1-098	Dawson-South Justice 115kV	Engineering and Procurement
AC1-099	Dawson-South Justice 115kV	Engineering and Procurement
AC1-189	Chinquapin-Everetts 230kV	Active
AC1-208	Cox-Whitakers 115kV	Active

Queue Number	Project Name	Status
AC2-084	Dawson-South Justice 115kV	Active
AD1-023	Cashie-Trowbridge 230 kV	Active
AD1-057	Hornertown-Hathaway 230 kV	Active
AD1-076	Trowbridge 230 kV	Active
AD2-051	Earleys – Northampton 230kV	Active
AE1-026	Cashie 230 kV	Active
AE1-035	Earleys 230 kV	Engineering and Procurement
AE1-103	Holland-Union Camp 115 kV	Active
AE2-033	Clubhouse-Sappony 230 kV	Active
AE2-034	Mackeys 230 kV	Active
AE2-040	Sapony 34.5 kV	Active
AE2-044	Anaconda-Dunbar 115 kV	Active
AE2-147	Swamp 230 kV	Active
AE2-151	Earleys 34.5kV	Engineering and Procurement
AE2-260	Clubhouse 230 kV	Active
AE2-304	South Hertford 34 kV	Engineering and Procurement
AE2-346	Ahoskie 34.5 kV	Active
AF1-059	Brodnax-South Hill 115 kV	Active
AF1-082	Heartsease-Mayo Dunbar DP	Active
AF1-152	Swamp 230 kV	Active
AF1-236	Mackeys 230 kV	Active
AF1-266	Clubhouse-Sappony 230 kV	Active
AF1-292	Fields Crossroads 34.5 kV	Active
V4-068	Murphy's 34.5kV	In Service
W1-029	Winfall 230kV	In Service
Z1-036	WinFall-Chowan 230kV	Suspended
Z2-043	Kelford 34.5kV	In Service
Z2-044	Whitakers 34.5kV	In Service
Z2-088	Tarboro-Everetts 230kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P1-2: LN 2056-A	CONTINGENCY 'DVP_P1-2: LN 2056-A' OPEN BRANCH FROM BUS 313845 TO BUS 934330 CKT 1 /* 6HATHAWAY 230.00 - AD1-057 TAP 230.00 END
DVP_P1-2: LN 2181	CONTINGENCY 'DVP_P1-2: LN 2181' OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00 OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-RMOUNT#4230.00 - 6NASH 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00 OPEN BUS 314591 /* ISLAND: 6NASH 230.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 2058	CONTINGENCY 'DVP_P1-2: LN 2058' OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00 OPEN BRANCH FROM BUS 313844 TO BUS 313845 CKT 2 /* 3HATHAWAY 115.00 - 6HATHAWAY 230.00 END
DVP_P1-2: LN 126	CONTINGENCY 'DVP_P1-2: LN 126' OPEN BRANCH FROM BUS 314568 TO BUS 314582 CKT 1 /* 3EARLEYS 115.00 - 3KELFORD 115.00 OPEN BRANCH FROM BUS 314582 TO BUS 314602 CKT 1 /* 3KELFORD 115.00 - 3SAMS HD 115.00 OPEN BRANCH FROM BUS 314602 TO BUS 314603 CKT 1 /* 3SAMS HD 115.00 - 3SCOT NK 115.00 OPEN BUS 314582 /* ISLAND: 3KELFORD 115.00 OPEN BUS 917332 /* ISLAND: Z2-043 E 115.00 OPEN BUS 314602 /* ISLAND: 3SAMS HD 115.00 END
DVP_P1-3: 6HATHAWAY-TX#1	CONTINGENCY 'DVP_P1-3: 6HATHAWAY-TX#1' OPEN BRANCH FROM BUS 313844 TO BUS 313845 CKT 1 /* 3HATHAWAY 115.00 - 6HATHAWAY 230.00 END
Base Case	
DVP_P4-2: 2058T2167	CONTINGENCY 'DVP_P4-2: 2058T2167' /* HATHAWAY 230 KV OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00 OPEN BRANCH FROM BUS 313844 TO BUS 313845 CKT 2 /* 3HATHAWAY 115.00 - 6HATHAWAY 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314564 CKT 1 /* 6HATHAWAY 230.00 - 6EDGECOM 230.00 END
DVP_P7-1: LN 2058-2181	CONTINGENCY 'DVP_P7-1: LN 2058-2181' /* . OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00 OPEN BRANCH FROM BUS 313844 TO BUS 313845 CKT 2 /* 3HATHAWAY 115.00 - 6HATHAWAY 230.00 OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00 OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-RMOUNT#4230.00 - 6NASH 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00 OPEN BUS 314591 /* ISLAND: 6NASH 230.00 END

12 Light Load Analysis

The Queue Project AF1-082 was evaluated as a 18.0 MW uprate to AC1-034 which is an injection tapping the Mayo Dunbar to Heartsease DP 115 kV line in the Dominion area. Project AF1-082 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-082 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

12.1 Light Load Deliverability

(Single or N-1 contingencies)

None

12.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies)

None

12.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

12.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

12.5 System Reinforcements

None

13 Short Circuit Analysis

No circuit breakers were identified as overdutied as part of this analysis.

14 Stability and Reactive Power

To be determined in the Facilities Study Phase.

15 Affected Systems

15.1 Duke Energy-Progress

Potential constraints were identified by PJM on the following Dominion – Duke Energy/Progress (DEP) tie lines. Enter into an Affected System Facilities Study agreement with Duke / Progress Energy (DEP) to determine what, if any, reinforcements are required on their system. The following facilities were identified in this report:

- Battleboro – Rocky Mt. 115 kV line

Attachment 1: One Line Diagram

